Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssw0rd'
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssw0rd” | base64
```

PowerShell Command: -

```
[Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("P@ssw0rd"))
```

```
PowerShell 7.5.4
PS C:\Windows\System32> cd C:\Users\HP\Desktop\COAS_Lab\Lab_8
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssw0rd'
secret/my-secret created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T06:09:44Z"
  name: my-secret
  namespace: default
  resourceVersion: "64156"
  uid: 047eb4de-cfe7-471a-9a0d-28c840f6dc19
type: Opaque
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> wsl
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar  3 11:41:40 IST 2026

  System load:  3.08                Processes:             39
  Usage of /:   0.6% of 1006.85GB   Users logged in:       1
  Memory usage: 19%                 IPv4 address for eth0: 172.22.242.177
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/root/.hushlogin file.
root@Shivang:/mnt/c/Users/HP/Desktop/COAS_Lab/Lab_8# echo -n “P@ssw0rd” | base64
4oCcUEBzc3cwcmTigJ0=
root@Shivang:/mnt/c/Users/HP/Desktop/COAS_Lab/Lab_8#
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> [Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("P@ssw0rd"))
UEBzc3cwcmQ=
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt
echo -n 'my-password' > ./password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> echo -n 'my-username' > ./username.txt
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> echo -n 'my-password' > ./password.txt
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
secret/my-file-secret created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQNCg==
  username: bXktdXNlcm5hbWUNCg==
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T06:14:27Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "64669"
  uid: 395a41a4-b9a2-499a-8f2f-37863a828ef8
type: Opaque
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-pa
ssword> --docker-email=<your-email> --docker-server=<your-docker-server>
secret/my-registry-secret created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJcdTAwM2N5b3VyLWRvY2tlci1zZXJ2ZXJcdTAwM2UiOnsidXNlcm5hbWUiOiJcdTAwM2N5b3VyLXVzZXJuYW1lXHUwMDNlIiwicGFzc3dvcmQiOiJcdTAwM2N
5b3VyLXBhc3N3b3JkXHUwMDNlIiwiZW1haWwiOiJcdTAwM2N5b3VyLWVtYWlsXHUwMDNlIiwiYXV0aCI6IlBIbHZkWEl0ZFhObGNtNWhiV1UrT2p4NWIzVnlMWEJoYzNOM2IzSmtQZz09In19fQ==
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T06:20:02Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "65276"
  uid: 578feb90-8d22-4db3-8f83-c06bcfa304a2
type: kubernetes.io/dockerconfigjson
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-secret.yaml
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```yaml
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f my-secret.yaml
secret/my-manifest-secret created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T06:09:44Z"
  name: my-secret
  namespace: default
  resourceVersion: "64156"
  uid: 047eb4de-cfe7-471a-9a0d-28c840f6dc19
type: Opaque
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

pod-with-secret.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
grep: The term 'grep' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> wsl
root@Shivang:/mnt/c/Users/HP/Desktop/COAS_Lab/Lab_8# kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
USERNAME=admin
PASSWORD=P@ssw0rd
root@Shivang:/mnt/c/Users/HP/Desktop/COAS_Lab/Lab_8#
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete secret my-secret
secret "my-secret" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete secret my-file-secret
secret "my-file-secret" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete secret my-registry-secret
secret "my-registry-secret" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete secret my-manifest-secret
secret "my-manifest-secret" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```